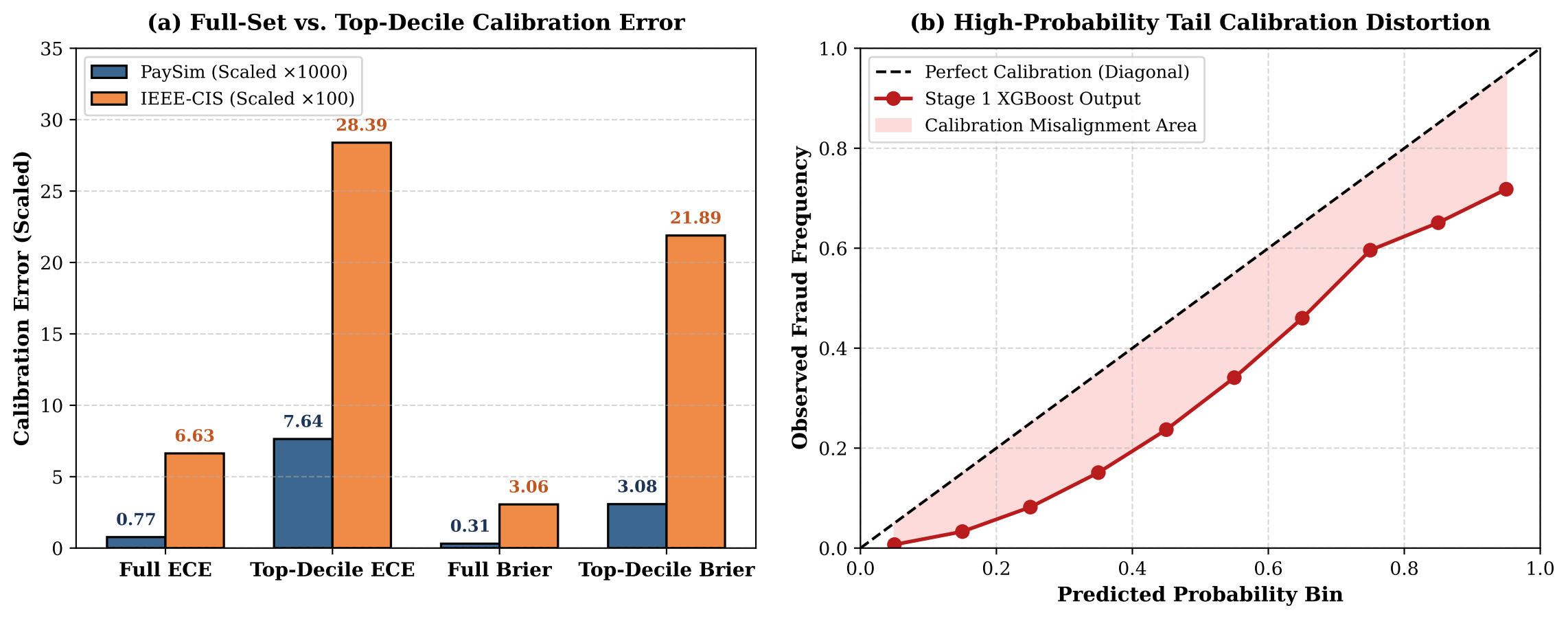
(a) Full-Set vs. Top-Decile Calibration Error
PaySim (Scaled ×1000)
IEEE-CIS (Scaled ×100)
Calibration Error (Scaled)
0
5
10
15
20
25
30
35
0.77
6.63
7.64
28.39
0.31
3.06
3.08
21.89
Full ECE
Top-Decile ECE
Full Brier
Top-Decile Brier
(b) High-Probability Tail Calibration Distortion
Perfect Calibration (Diagonal)
Stage 1 XGBoost Output
Calibration Misalignment Area
Observed Fraud Frequency
0.0
0.2
0.4
0.6
0.8
1.0
Predicted Probability Bin